Eric Xu

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Education

University of California, Santa Cruz — Santa Cruz, CA
M.S. in Scientific Computing and Applied Mathematics — *2026 – 2027*
B.S. in Applied Mathematics, Highest Honors in the Major | GPA 3.77/4.00 | Dean's Honors (4 quarters) — *2024 – 2026*

- **Graduate coursework:** Scientific Machine Learning, Applied Dynamical Systems, Machine Learning and AI in Genomics

Experience

TACS Lab, UC Santa Cruz — Santa Cruz, CA
Researcher, B.S. and M.S. (Advisor: Prof. Ashesh Chattopadhyay) — *Sep 2025 – Present*

- Develop an AI ensemble method that combines machine-learned weather models with rare-event resampling to forecast heat waves and cold outbreaks weeks ahead.
- Test it on past heat waves and cold outbreaks across the US, with extreme precipitation planned next.

Vayuh.ai
ML Researcher (part-time) — *Jul 2026 – Present*

- Run multi-GPU inference and benchmarking experiments for subseasonal extreme-weather forecasts on H100 nodes.
- Cut the GPU cost of extreme-event forecasts by benchmarking AI weather models and moving the ensemble to a faster one.
- Fixed JAX compile caching for GenCast so forecast runs stop recompiling the model.

Projects

Subseasonal Extreme Weather Forecasting | *JAX, SLURM, PBS* — Jun 2026 – Present

- Built a batch pipeline that starts AI weather-model ensembles weeks before past heat waves and cold outbreaks and checks whether they capture the observed extreme.
- Extending the tests to extreme precipitation events.

Council of Sumer | *Python, FastAPI, AGNTCY SLIM, OpenTelemetry* — Oct 2026

- Built a hackathon system (team of 3) where LLM agents with conflicting goals run as separate processes, debate over AGNTCY SLIM, and vote under runoff and human tie-break rules.
- Co-wrote the deliberation engine and wrote the benchmark that compares the council with a single LLM.

Cross-Species Transfer Learning with DeepSEA | *PyTorch, scikit-learn* — Jun 2026

- Fine-tuned the human-trained DeepSEA CNN on mouse ENCODE DNA windows. It beat training from scratch on most chromatin features and converged in fewer epochs.
- Showed that zero-shot transfer to mouse scores near chance, so the model needs fine-tuning.

4D-Var Data Assimilation with a Learned Inverse Operator | *PyTorch, L-BFGS* — Apr – May 2026

- Reimplemented in PyTorch a Google Research 4D-Var method that uses a learned CNN inverse observation operator. On Lorenz-96 it beat standard 4D-Var on state-estimation error.

Teaching Experience

Baskin School of Engineering, UC Santa Cruz — Santa Cruz, CA
Teaching Assistant: Differential Calculus for Economics, Ordinary Differential Equations — *Sep 2026 – Present*

- Lead weekly discussion sections and office hours, and grade assignments.

Tutor and Reader: Linear Algebra, Discrete Mathematics, Calculus II — *Sep 2025 – Jun 2026*

- Tutored for 3 quarters and graded for 1 quarter.

Skills

Mathematics: Linear algebra, ODEs and PDEs, dynamical systems, numerical methods, probability and statistics
Languages: Python, C++, MATLAB, R, Bash, RISC-V assembly
ML and Scientific Computing: PyTorch, JAX, NumPy, scikit-learn, Triton (GPU kernels)
HPC and Tools: SLURM, PBS, NCAR Derecho, Linux, Git, SSH